Product of the roots $(\alpha^2\beta)\times(\beta^2\alpha) = \alpha^3\beta^3 = (\alpha\beta)^3 = \left(-\frac{1}{2}\right)^3 = -\frac{1}{8}$

The required equation is $x^2 - (\text{Sum of the roots})x + (\text{Product of the roots}) = 0$

$$x^2 - \left(-\frac{1}{4}\right)x - \frac{1}{8} = 0 \;\Rightarrow\; 8x^2 + 2x - 1 = 0$$

(iii) $2\alpha + \beta,\ 2\beta + \alpha$

Sum of the roots $2\alpha + \beta + 2\beta + \alpha = 3(\alpha + \beta) = 3\left(\frac{1}{2}\right) = \frac{3}{2}$

Product of the roots $= (2\alpha + \beta)(2\beta + \alpha) = 4\alpha\beta + 2\alpha^2 + 2\beta^2 + \alpha\beta$

$$= 5\alpha\beta + 2(\alpha^2 + \beta^2) = 5\alpha\beta + 2\left[(\alpha + \beta)^2 - 2\alpha\beta\right]$$

$$= 5\left(-\frac{1}{2}\right) + 2\left[\frac{1}{4} - 2\times -\frac{1}{2}\right] = 0$$

The required equation is $x^2 - (\text{Sum of the roots})x + (\text{Product of the roots}) = 0$

$$x^2 - \frac{3}{2}x + 0 = 0 \;\Rightarrow\; 2x^2 - 3x = 0$$

Exercise 3.14

1. Write each of the following expression in terms of $\alpha + \beta$ and $\alpha\beta$.

   (i) $\frac{\alpha}{3\beta} + \frac{\beta}{3\alpha}$ (ii) $\frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha}$ (iii) $(3\alpha - 1)(3\beta - 1)$ (iv) $\frac{\alpha + 3}{\beta} + \frac{\beta + 3}{\alpha}$

2. The roots of the equation $2x^2 - 7x + 5 = 0$ are α and β. Without solving for the roots, find

   (i) $\frac{1}{\alpha} + \frac{1}{\beta}$ (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (iii) $\frac{\alpha + 2}{\beta + 2} + \frac{\beta + 2}{\alpha + 2}$

3. The roots of the equation $x^2 + 6x - 4 = 0$ are α, β. Find the quadratic equation whose roots are

   (i) α^2 and β^2 (ii) $\frac{2}{\alpha}$ and $\frac{2}{\beta}$ (iii) $\alpha^2\beta$ and $\beta^2\alpha$

4. If α, β are the roots of $7x^2 + ax + 2 = 0$ and if $\beta - \alpha = \frac{-13}{7}$. Find the values of a.

5. If one root of the equation $2y^2 - ay + 64 = 0$ is twice the other then find the values of a.

6. If one root of the equation $3x^2 + kx + 81 = 0$ (having real roots) is the square of the other then find k.

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